

Practice Session 4 Answers

Part 1: Sampling Distributions

1.1 Introduction

Recall that a “sample” is a subset of a population. We can use samples to calculate “statistics”, such as the sample mean, median, and standard deviation. If we were to **repeatedly** sample from the same population and calculate the **mean** of each sample, we would obtain what is known as the **sampling distribution** of the **sample mean**. If we were to instead calculate a **proportion** from each sample, we would obtain the sampling distribution of the **sample proportion**.

In general, a sampling distribution is the distribution of a **statistic** across many samples. This is in contrast to the population distribution, which is the distribution of the values of all individuals (or sampling units) in the population.

1.2 Sampling Distribution Terminology

Match each term with its definition.

Terms:

- (a) s
- (b) σ
- (c) Standard Error (SE)
- (d) x
- (e) $\bar{x}$
- (f) μ

Definitions:

- Standard deviation of a population
- Individual unit of the population
- Mean of a population
- Standard deviation of a sample
- Sample mean
- Standard deviation of a sampling distribution

Answers

- (a) s = Standard deviation of a sample
- (b) σ = Standard deviation of a population
- (c) Standard Error (SE) = Standard deviation of a sampling distribution
- (d) x = Individual unit of the population
- (e) $\bar{x}$ = Sample mean
- (f) μ = Mean of a population

1.3 Generating Sampling Distributions

Let's generate a sampling distribution using polling data on the president's approval rating. Make sure to load the `library(SDS1000)` to have access to the functions for this exercise.

1. Use the `rapprovals()` function to obtain a sample of $n = 1000$ individual approval ratings. Use the `get_proportion()` function to obtain the proportion of voters in this sample that approve of the president.

```
# Load the SDS1000 package
library(SDS1000)

# Setting the seed so that we all get the same "random sample"
set.seed(1000)

# Get a random sample of 1000 fictional voters' opinions of the president
approval_sample <- rapprovals(1000)

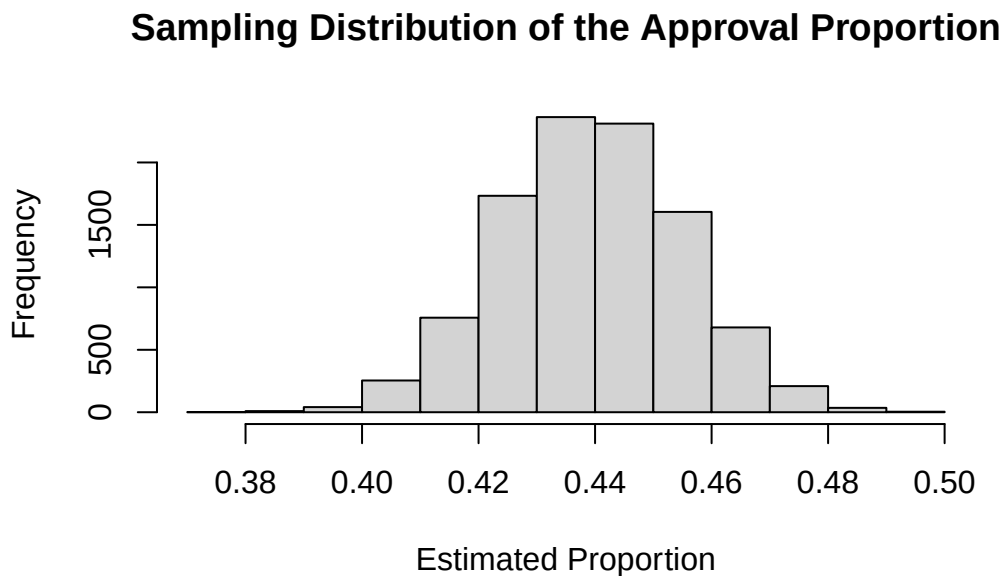
# Get the proportion that approves
p_hat <- get_proportion(approval_sample, "approve")
p_hat
```

```
approve
0.444
```

2. Next, use the `do_it()` function from class to generate 10,000 samples, and repeat the above steps to obtain the proportion for each sample. Assign this vector of proportions to the variable `polling_distribution`, and create a histogram of it. Make sure to have an appropriate title.

Challenge: What do values on the x-axis of this histogram correspond to? How does this differ from histograms that we've previously encountered?

```
polling_distribution <- do_it(10000) * {  
  
  curr_sample <- rapprovals(1000)  
  get_proportion(curr_sample, "approve")  
  
}  
  
hist(polling_distribution, main = "Sampling Distribution of the Approval Proportion",  
     xlab = "Estimated Proportion")
```



Answer: Each value on the x-axis is a **sample proportion** — a single statistic computed from an entire sample of 1000 voters — not the response of any one individual. The histograms we've made before plotted the values of individual observations (one value per person or unit), so they showed the distribution of the data. Here each observation is a whole sample, so the histogram shows how the statistic varies from sample to sample: it is a histogram of a sampling distribution, not of a population or a single sample.

3. Finally, calculate the mean and standard deviation of your collection of sample proportions. What is another name for the standard deviation?

```
mean(polling_distribution)
```

```
[1] 0.4398881
```

```
sd(polling_distribution)
```

```
[1] 0.01563533
```

Answer: The standard deviation of a sampling distribution is called the **standard error (SE)**. Here the SE is about 0.016, which matches the theoretical value $\sqrt{0.44 \times 0.56/1000} \approx 0.0157$.

1.4 Properties of Sampling Distributions

Answer the following True/False questions relating to sampling distributions.

- (a) True/False: As the sample size n decreases, the standard error of a sampling distribution also decreases.
- (b) True/False: The mean of a sampling distribution is always **greater than** the mean of the population.
- (c) True/False: For sample sizes larger than 1, the standard deviation of the sampling distribution of the sample mean is **smaller than** the standard deviation of the population.
- (d) True/False: Assuming that you have an unbiased sampling procedure with a sufficient sample size, the sampling distribution of the sample mean is approximately normally distributed.
- (e) True/False: It is generally better to have a larger standard error than a smaller one.

Answers

- (a) False. The SE increases as n decreases; for the sample mean, $SE = \sigma/\sqrt{n}$, so a smaller n always gives a larger SE.
- (b) False. Assuming the sampling process is unbiased and random, the mean of the sampling distribution will be equal to the mean of the population.

- (c) True. For the sample mean, $SE = \sigma/\sqrt{n}$, which is smaller than σ whenever $n > 1$. Averaging over many individuals smooths out individual variation, so sample means vary less than individual values do.
- (d) True. By the Central Limit Theorem, the sampling distribution of the sample mean (and of a sample proportion) is approximately normal, assuming that your sampling procedure is unbiased and has a sufficient n . This does not hold for every statistic — for example, the sampling distribution of a sample maximum or a sample variance need not be normal.
- (e) False. Smaller standard errors are typically preferred, since they mean our estimates vary less from sample to sample.

Part 2: More Practice on Sampling Distributions

2.1 Practice: Baseball Salaries

For this section, we will use the `BaseballSalaries2015` data set from the `Lock5Data` library. Salaries are recorded in millions of dollars.

1. Using the `sample()` function, obtain a sample of $n = 100$ baseball player salaries. See the `help` page for `sample()` for details.

```
library(SDS1000)
library(Lock5Data)
set.seed(1000)

salary_sample <- sample(BaseballSalaries2015$Salary, size = 100)
```

2. Now, using the `do_it()` function, repeat this process 10,000 times, and calculate the mean each time. Compare the histogram of these sample means to a histogram of the individual salaries in the population.

```
salary_distribution <- do_it(10000) * {

  curr_sample <- sample(BaseballSalaries2015$Salary, size = 100)
  mean(curr_sample)

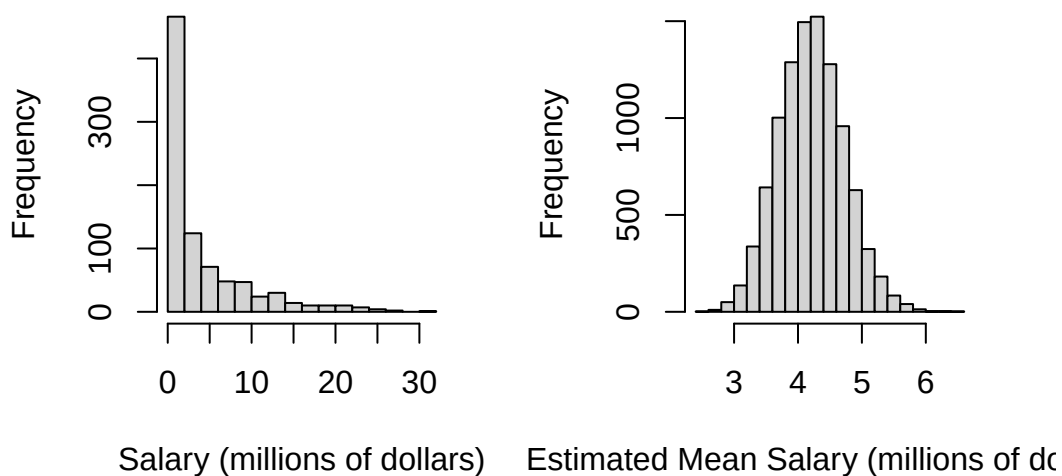
}
```

```
par(mfrow = c(1, 2))

hist(BaseballSalaries2015$Salary, main = "Population: Individual Salaries",
     xlab = "Salary (millions of dollars)")

hist(salary_distribution, main = "Sampling Distribution of the Mean",
     xlab = "Estimated Mean Salary (millions of dollars)")
```

Population: Individual Salary Sampling Distribution of the Mean



Notice that the individual salaries are strongly right-skewed, yet the sampling distribution of the mean is roughly symmetric and bell-shaped. The population does not have to be normal for the sampling distribution of the sample mean to be approximately normal. Notice also that the x-axes measure different things: the left histogram shows individual players' salaries, while the right histogram shows the means of samples of 100 players (see the Challenge question in 1.3).

- Using your original sample of 100 salaries from question 1.), calculate a point estimate of the mean salary.

```
point_estimate <- mean(salary_sample)
point_estimate
```

```
[1] 3.44262
```

4. Calculate the standard deviation of your sampling distribution from question 2.) to find the standard error.

```
se <- sd(salary_distribution)
se
```

```
[1] 0.5166305
```

Part 3: Confidence Intervals

3.1 Confidence Interval Terminology

Match each term with its definition.

Terms:

- (a) Point Estimate
- (b) Interval Estimate
- (c) Confidence Interval
- (d) Confidence Level

Definitions:

- A particular kind of interval estimate, produced by a method such that, over repeated samples, a specified percentage of the intervals produced (the confidence level) will contain the population parameter.
- The percentage of confidence intervals that will contain the population parameter of interest over repeated sampling. For example, a 95% confidence level means that, in the long run, 95% of the confidence intervals generated by the method will contain the population parameter of interest.
- A single sample statistic that we use to estimate a population parameter. The sample mean is an example, as we use it to estimate the population mean.
- A range of plausible values for a population parameter, rather than a single number. It does not by itself carry a stated confidence level. Many interval estimates take the form point estimate $\pm$ margin of error, in which case the point estimate sits at the center.

Answers

- (a) Point Estimate = A single sample statistic that we use to estimate a population parameter.
- (b) Interval Estimate = A range of plausible values for a population parameter, rather than a single number.
- (c) Confidence Interval = A particular kind of interval estimate, produced by a method such that, over repeated samples, a specified percentage of the intervals will contain the population parameter.
- (d) Confidence Level = The percentage of confidence intervals that will contain the population parameter over repeated sampling.

3.2 Confidence Interval for Baseball Salaries

For this section, we will use the `BaseballSalaries2015` data set from the `Lock5Data` library.

1.) Using the `sample()` function, obtain a sample of $n = 100$ baseball player salaries. Make sure to sample **without replacement**. See the `help` page for `sample()` for details.

```
library(Lock5Data)
set.seed(1000)
salary_sample <- sample(BaseballSalaries2015$Salary, size = 100, replace = F)
```

2.) Now, using the `do_it()` function, repeat this process $n = 10,000$ times, and calculate the mean each time.

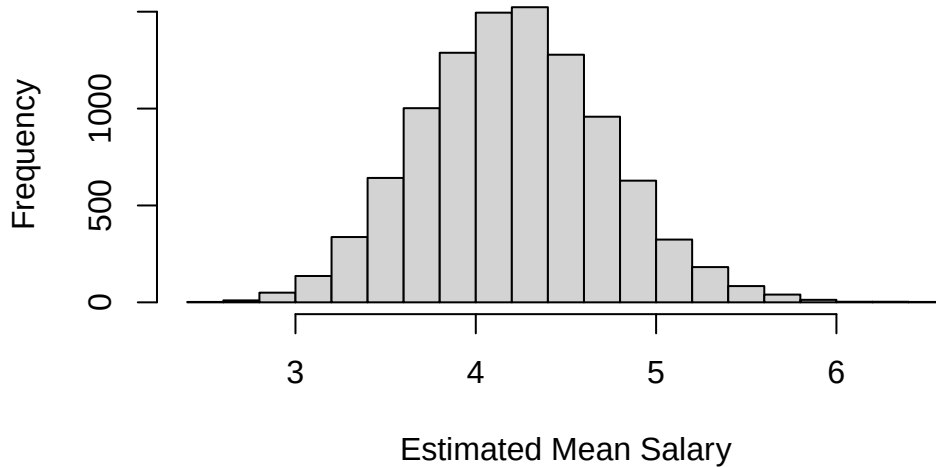
```
salary_distribution <- do_it(10000) * {

  curr_sample <- sample(BaseballSalaries2015$Salary, size = 100, replace = F)
  mean(curr_sample)

}

hist(salary_distribution, main = "Distribution of Salary Samples",
      xlab = "Estimated Mean Salary")
```

Distribution of Salary Samples



3.) Using your original sample of 100 salaries from part 1.), calculate a point estimate of the mean salary.

```
mu_estimate = mean(salary_distribution)
```

4.) Calculate the standard deviation of your sampling distribution from part 2.) to estimate the standard error.

```
se <- sd(salary_distribution)
```

5.) Using your answers from part 3.) and 4.), generate a 95% interval estimate for the mean salary of baseball players. The interval estimate is of the form:

$$\text{Point Estimate} \pm 1.96 \cdot \text{Standard Error}$$

.

```
int_estimate <- c(mu_estimate - 1.96 * se, mu_estimate + 1.96 * se)
int_estimate
```

```
[1] 3.199773 5.224964
```

```
# or you can calculated it
# int_estimate <- mu_estimate + c( - 1.96, 1.96) * se
```

6.) Interpret your interval estimate.

NOTE: Answers may vary due to randomness in sampling

“The true average salary of baseball players for the 2015 season is likely to lie in the range [Lower-Bound, Upper-Bound], since 95% of the time we use this method the parameter is in such intervals.”

7.) Compare your interval estimates to the actual mean salary from the entire data set. Comment on what you notice.

```
mu <- mean(BaseballSalaries2015$Salary)
sigma <- sd(BaseballSalaries2015$Salary)
mu
```

```
[1] 4.214702
```

We see that our interval contains the true mean salary.